

Varnan Rathod

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EDUCATION

Shri Ramdeobaba College of Engineering and Management
B.Tech. in Computer Science and Engineering (Honors in AI/ML): 9.26 CGPA

Nagpur, MH
Aug 2023 – May 2027

School of Scholars, Wanadongri
HSC: 92%

Nagpur, MH
Jul 2021 – May 2023

RESEARCH EXPERIENCE

Research Intern
Indian Institute of Technology (IIT) Roorkee

September 2026 – Present
Roorkee, UK (part time)

Faculty Mentor: Dr. Gaurav Kumar Nayak, Mehta School of DS & AI

- Researching novel methodologies at the intersection of computer vision (CV) and natural language processing (NLP).

Undergraduate Researcher
Ramdeobaba University

Jul 2026 – Present
Nagpur, MH (part time)

Faculty Mentor: Dr. Khushboo Khurana, Department of CSE

- Extending a transformer-based object detection architecture for small object detection in aerial imagery.
- Working with modifications inspired by signal-processing and studying the effect of various loss functions on the same. Ablation study in progress.

Research Intern
Visvesvaraya National Institute of Technology (VNIT)

Dec 2025 – August 2026
Nagpur, MH (part time)

Faculty Mentor: Dr. Praveen Kumar Verma, Department of CSE

- Experimented, tested, and integrated open-source state-of-the-art face recognition models and optimal data storage and retrieval functionalities for a targeted large-scale attendance monitoring full stack application.
- Training, experimenting with, and evaluating RNN-based multimodal sequence models across diverse input signals and modalities for ongoing research.

Undergraduate Researcher
Ramdeobaba University

Aug 2025 – Oct 2025
Nagpur, MH (part time)

Faculty Mentor: Dr. Khushboo Khurana

- Conducted a comparative study of Transformer-based object detection models, including DiffusionDet and DEIM.
- Analyzed architectural trade-offs and scene understanding mechanisms in DETR-style detectors.
- Used a curated dataset for a particular use case and benchmarked models using mAP across various object scales and confidence thresholds.
- Fine-tuned selected models, achieving state-of-the-art (SOTA) bounding box detection results on the dataset.

Research & Development Intern
Ramdeobaba University

May 2025 – Feb 2026
Nagpur, MH (part time)

Project: Development of Coal Quality Exploration Technique using CNNs and Hyperspectral Imaging

Sponsored by: Ministry of Coal, Government of India

Guide: Principal & Prof. M. B. Chandak | **Coordinator:** Dr. Abhijeet Raipurkar

- Implemented a CNN-based regression pipeline for coal quality assessment using hyperspectral image datacubes.
- Constructed and preprocessed a novel hyperspectral dataset acquired from a newly commissioned imaging sensor.
- Analyzed latent correlations across spectral bands to inform model adaptation and optimization.
- Fine-tuned a pretrained CNN architecture on a dataset approximately 3% the size of the original training corpus.
- Achieved a 72% reduction in MAE relative to baseline.

PAPERS & ACADEMIC SERVICE

- **[Paper] Utilizing DETR and Diffusion-Based Models for Underwater Debris Detection and Recognition** - Presented at *IEEE Recent Advances in Electronics, Communication, Computing, Automation, and Power (RECCAP) 2026*, IIT Palakkad; awaiting publication and indexing results
- **[Reviewer]** - *IEEE Guwahati Subsection Conference (GCon) 2026*, at IIT Guwahati (*voluntary peer reviewer, selected as undergraduate*)
- **[Reviewer]** - *IEEE Gujarat Subsection Conference (GSCon) 2027*, at SVNIT Surat (*voluntary peer reviewer, selected as undergraduate*)

TESTS & SCORES

- **GATE CS 2026:** AIR 2472, **98.82%ile**, Score: 625
- **JEE Advanced:** Secured an All India Rank (AIR) of **22704**
- **JEE Main (B.Tech):** Achieved a percentile of **95.72%ile**
- **MHT-CET:** Secured a percentile of **98.77%ile**
- **Bebras Computing Challenge 2022:** Ranked **9th** nationally out of 6,369 participants

RELEVANT COURSES

Course Code	Course Name	Grade
CATH 6100	Generative Adversarial Networks	AA
CATH 5100	AI for Vision and Navigation	AA
CATH 3100	Computational Statistics For Data Science	AB
CST 5003	Artificial Intelligence	AA
CST 3002	Theory Of Computation	AB
CST 3001	Data Structures	AB
MAT 4001	Linear Algebra	AA
MAT 3001	Probability and Statistics	AA
MAT 2001	Discrete Mathematics	AA
MAT 1001	Calculus	AA

ACHIEVEMENTS & ROLES

- **Finalist:** Datathon, ARTIMAS'25, PCCOE Pune
- **Grand Finalist:** Smart India Hackathon (SIH) 2024
- **Executive Tech Member:** GFG RBU Chapter
- **Academic Mentor (2024-2025 & 2025-2026):** RBU
- **Web Development Member:** Technical Club, RBU
- **City Rank 2:** CSIR NEERI Science Quiz 2023

TECHNICAL SKILLS

Languages: Python, Java, SQL, C

Frameworks: TensorFlow, Keras, PyTorch

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, LaTeX

Libraries: NumPy, Pandas, Seaborn, Matplotlib, Scikit-Learn, SciPy, XGBoost, SHAP

Databases: PostgreSQL, MongoDB